Structure-Aware Prediction of PROTAC-Mediated Protein Degradability via Graph Neural Networks

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Abstract

Proteolysis-targeting chimeras (PROTACs) can selectively degrade disease-causing proteins, yet predicting which targets are amenable to degradation remains a critical bottleneck: existing computational methods require the complete PROTAC molecular structure, information unavailable before synthesis. We present DegradoMap, a graph neural network that predicts PROTAC-mediated degradability from protein structure and E3 ligase identity alone—the minimal information available at the target selection stage. The model encodes biophysical priors through lysine-weighted graph pooling with per-protein normalization, models protein–E3 compatibility via cross-attention, and integrates cellular context from the Cancer Dependency Map. On the PROTAC-8K benchmark (3,101 samples, 155 targets, 10 E3 ligases), DegradoMap achieves 0.646 $\pm$0.124 AUROC on target-unseen evaluation (best seed: 0.7449) and 0.811 AUROC on CRBN$\rightarrow$VHL E3-unseen transfer, outperforming GNN and machine learning baselines. The model additionally recommends optimal E3 ligases with 74% Hit@3 accuracy. Two findings carry broader implications: E(3)-equivariant architectures underperform the simpler invariant design for this scalar prediction task, and ESM-2 embeddings improve peak performance only with careful regularization—naive integration fails. DegradoMap provides pre-synthesis computational guidance for degradability assessment; its well-calibrated confidence scores (ECE = 0.029, target-unseen) enable practitioners to prioritize high-confidence predictions for experimental follow-up. However, the high seed variance (std = 0.124) and limited E3 coverage require ensembling for reliable deployment.

CCS Concepts

• **Computing methodologies** → **Neural networks**; • **Applied computing** → **Bioinformatics**; *Computational biology*.

Keywords

PROTAC, targeted protein degradation, graph neural network, drug discovery, ubiquitin–proteasome system, AlphaFold

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1 Introduction

Proteolysis-targeting chimeras (PROTACs) invert the logic of drug design: rather than occupying a protein's active site, they recruit the cell's ubiquitin–proteasome machinery to destroy the protein entirely [1–3]. A PROTAC bridges two binding events—its warhead engages the target protein while its E3 ligand recruits an E3 ubiquitin ligase—and the induced proximity triggers polyubiquitination of surface lysine residues, marking the target for proteasomal degradation [4]. Because this mechanism is catalytic, a single PROTAC molecule degrades multiple target copies, enabling efficacy at sub-stoichiometric concentrations and access to proteins that lack traditional drug-binding pockets.

With over 20 PROTACs now in clinical trials [1], the modality's therapeutic promise is clear. Yet development faces a critical bottleneck at its earliest stage: *target selection*. Is a given disease-relevant protein amenable to PROTAC-mediated degradation? Answering this currently requires synthesizing candidate PROTACs and conducting cell-based degradation assays—an expensive process with high failure rates. Computational prediction of degradability *before* PROTAC synthesis would dramatically accelerate the pipeline.

Limitations of existing methods. Existing computational approaches all require the PROTAC molecular structure as input. DeepPROTACs [5] operates on voxelized binding pockets requiring complete PROTAC, target, and E3 coordinates. Ribes et al. [6] combine protein embeddings with PROTAC SMILES fingerprints. DegradeMaster [8] achieves 0.856 AUROC using full PROTAC molecular descriptors alongside protein features. While valuable for optimizing existing PROTACs, these methods cannot serve target selection because the PROTAC is unavailable at that stage. MAPD [7] addresses target assessment from protein-intrinsic features but does not leverage AlphaFold structural information [9].

Our approach. We introduce DegradoMap, a graph neural network that predicts protein degradability from *minimal inputs*: a protein structure (from AlphaFold, requiring only a UniProt ID) and E3 ligase identity (e.g., "CRBN" or "VHL"). The design reflects the biophysics of ubiquitin transfer, which requires (1) surface-accessible lysine residues, (2) geometric compatibility between target and E3 ligase for ternary complex formation, and (3) sufficient cellular E3 expression and proteasome capacity. DegradoMap addresses these requirements through three modules and a gated fusion mechanism:

(1) **Structure–Ubiquitination Graph (SUG) encoder**: An invariant message-passing GNN with lysine-weighted

pooling that focuses representation on ubiquitination-relevant residues, incorporating ESM-2 embeddings [14], structural properties, and known ubiquitination site annotations from PhosphoSitePlus.

(2) **E3 compatibility module**: Bidirectional cross-attention modeling compatibility between protein structure and E3 ligase identity, enabling generalization across E3 ligases.

(3) **Cellular context encoder**: A residual MLP processing 59 tissue-specific features from the Cancer Dependency Map [21].

(4) **Gated fusion**: Learned gating that dynamically weights module outputs for multi-task prediction of binary degradability and continuous efficiency.

Evaluation and contributions. On the PROTAC-8K benchmark [8] (3,101 samples, 155 targets, 10 E3 ligases), DegradoMap achieves a 3-seed mean of 0.646 ± 0.124 and 6-seed mean of 0.603 ± 0.097 AUROC on target-unseen evaluation (best seed: 0.7449), 0.811 AUROC [0.785, 0.836] on CRBN→VHL E3-unseen transfer, and 74% Hit@3 for E3 recommendation. We note the 6-seed improvement over gradient boosting baseline (0.607) is not statistically significant ($p = 0.556$), and VHL-targeted proteins fail below random (0.396 AUROC)—both limitations prominently acknowledged. Two architectural findings carry implications beyond PROTAC prediction: E(3)-equivariant architectures underperform the invariant design for scalar property prediction, and ESM-2 embeddings require task-specific regularization to improve performance. Our principal contributions include a structure-based pre-synthesis degradability predictor requiring only a UniProt ID and E3 ligase name, a lysine-weighted pooling mechanism encoding ubiquitin transfer biology, a cross-attention E3 compatibility module, well-calibrated confidence scores (ECE = 0.029) enabling threshold-based screening, and rigorous multi-seed evaluation revealing variance characteristics critical for deployment planning.

2 Related Work

PROTAC prediction methods. Computational approaches to PROTAC modeling vary significantly in their input requirements and intended use cases, as summarized in Table 1. DeepPROTACs [5] uses 3D CNNs on binding pocket voxelizations, requiring the complete PROTAC molecule plus target and E3 binding pockets in mol2 format—information available only after PROTAC synthesis and structural characterization. Ribes et al. [6] combine pretrained protein embeddings with molecular fingerprints, requiring target sequence and PROTAC SMILES, which presupposes PROTAC design. MAPD [7] predicts degradation tractability from protein-intrinsic features using machine learning; while it shares our target-assessment framing, it does not leverage structural information. DegradeMaster [8] achieves 0.856 AUROC using full PROTAC molecular descriptors alongside target features, representing the state-of-the-art for PROTAC optimization but addressing a fundamentally different problem than target selection. DegradoMap uniquely combines structural awareness with minimal input requirements, filling the gap between sequence-only assessment (MAPD) and full-PROTAC optimization (DeepPROTACs, DegradeMaster).

Table 1: Input modality comparison of PROTAC prediction methods. DegradoMap requires only protein structure and E3 identity, enabling use before PROTAC synthesis.

Method	Protein	PROTAC	E3	Use Case
DeepPROTACs	Pocket (3D)	Full mol.	Pocket (3D)	Optimization
Ribes et al.	Sequence	SMILES	–	General
MAPD	Features	–	–	Assessment
DegradeMaster	Features	Descriptors	Features	Optimization
DegradoMap	**Structure**	–	**Identity**	**Assessment**

Geometric deep learning for proteins. GNNs operating on protein structures have achieved strong performance across diverse tasks including function prediction, binding site identification, and property estimation. GVP-GNN [10] maintains SO(3)-equivariance using geometric vector perceptrons that jointly process scalar and vector features, enabling the model to reason about directional relationships. GearNet [11] employs relational graph convolutions with multiple edge types (sequential bonds, spatial contacts) for protein representation pre-training. SchNet [12] introduces continuous-filter convolutions with radial basis function expansions, originally designed for quantum chemistry but applicable to protein graphs. EGNN [13] achieves E(n)-equivariance through simple coordinate updates alongside feature updates, offering an elegant and computationally efficient approach. We evaluate SchNet and EGNN as structural baselines and find that while they perform competitively on random splits, they fail to generalize to unseen targets—suggesting that task-specific inductive biases (lysine-aware pooling, E3 compatibility modeling) are more important than architectural sophistication for degradability prediction.

Protein language models. ESM-2 [14], ProtTrans [15], and ESM-IF [16] learn rich representations from evolutionary sequences. PROTAC-mediated degradation—an artificial process—is not directly predicted by evolutionary features alone (ESM-2 alone: 0.534 AUROC; Section 4.5), but carefully integrated ESM-2 embeddings boost peak performance to 0.7449 AUROC, providing complementary signal alongside structural features.

Structure-based protein function prediction. DeepFRI [24] predicts Gene Ontology terms from contact maps using graph convolutions, demonstrating that protein structure encodes functional information beyond sequence. AlphaFold [9] has made high-quality structural predictions universally available, enabling structure-based ML at scale. Our work leverages both advances: we use AlphaFold-predicted structures as inputs and GNN-based structure processing for degradability prediction. However, degradability prediction differs fundamentally from function prediction in that it involves an *intermolecular* interaction (protein–PROTAC–E3 ternary complex) rather than intrinsic molecular properties.

Ubiquitin system modeling. UbiBrowser [17] and DeepUbi [18] predict natural ubiquitination sites, where E3 specificity is determined by evolved degrons. PROTAC-induced degradation bypasses this: the PROTAC bridges target and E3 artificially, so *surface lysine accessibility*—not natural specificity—determines degradability. This motivates our structure-first approach; natural ubiquitination data could provide pre-training signal for lysine representations.

3 Methods

3.1 Problem Formulation

We formulate PROTAC-mediated degradability prediction as a binary classification problem. Given a protein graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X}, \mathbf{C})$ with residue nodes $\mathcal{V}$, spatial edges $\mathcal{E}$, node features $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times d}$ (d=1,285 with ESM-2; d=28 without), and Cα coordinates $\mathbf{C}$, together with E3 ligase identity e, we predict:

$$\hat{y} = f_\theta(\mathcal{G}, e) = P(\text{degradable} \mid \mathcal{G}, e) \in [0, 1] \tag{1}$$

This formulation explicitly excludes the PROTAC molecular structure, reflecting our target-assessment use case where the PROTAC has not yet been designed. The protein graph is constructed as a radius graph with cutoff $r = 8\,\text{Å}$: $\mathcal{E} = \{(i, j) : \|\mathbf{c}_i - \mathbf{c}_j\|_2 < r,\ i \neq j\}$, connecting residues that are spatially proximal in 3D. This captures local tertiary structure contacts, surface accessibility patterns, and potential protein-protein interaction interfaces relevant to ternary complex formation.

Figure 1 illustrates the overall architecture.

3.2 Module A: Structure–Ubiquitination Graph (SUG)

The SUG module is the core component of DegradoMap, processing the protein structure graph through invariant message passing. We design this module around three biophysical principles: (1) local structural context determines surface accessibility, (2) lysine residues are the exclusive sites of ubiquitin attachment, and (3) protein size should not bias predictions. Node features can optionally include ESM-2 protein language model embeddings (1,280-dim per residue), which provide rich evolutionary and contextual representations when available (Section 4.5).

Message passing. At each layer $l \in \{1, \ldots, L\}$, node representations are updated via:

$$\mathbf{m}_{ij}^{(l)} = \phi_{\text{msg}}\left(\mathbf{h}_i^{(l)} \| \mathbf{h}_j^{(l)} \| d_{ij}\right) \tag{2}$$

$$\mathbf{h}_i^{(l+1)} = \phi_{\text{upd}}\left(\mathbf{h}_i^{(l)}, \sum_{j \in \mathcal{N}(i)} \mathbf{m}_{ij}^{(l)}\right) \tag{3}$$

where $\|$ denotes concatenation, $d_{ij} = \|\mathbf{c}_i - \mathbf{c}_j\|_2$ is the Euclidean distance between Cα atoms, ϕ_{msg} and ϕ_{upd} are 2-layer MLPs with LayerNorm and ReLU activations, and $\mathcal{N}(i) = \{j : d_{ij} < r\}$ denotes the spatial neighbors within radius $r = 8\,\text{Å}$. By using the scalar distance d_{ij} rather than the coordinate difference vector $\mathbf{c}_j - \mathbf{c}_i$, the message function is inherently invariant to rotation and translation—a property we validate empirically in Section 4.5. We use $L = 4$ message-passing layers with hidden dimension 128 and output dimension 64.

Lysine-weighted pooling. Ubiquitin is covalently conjugated to the ε-amino group of lysine side chains via an isopeptide bond. Therefore, degradation efficiency depends critically on the accessibility and spatial distribution of surface lysines relative to the E3 ligase binding interface. We incorporate this domain knowledge through a lysine-weighted global pooling mechanism:

$$\mathbf{h}_{\text{prot}} = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} w_i \cdot \mathbf{h}_i^{(L)} \tag{4}$$

where the weights $w_i = \text{softmax}_{i \in \mathcal{V}}(\alpha \cdot \mathbb{1}[\text{Lys}_i])$ are computed via *per-protein* softmax normalization over a learned scalar parameter α that controls the strength of lysine upweighting.

Two design choices are critical. First, the **per-protein softmax** (normalizing over residues within a single protein, not across the batch) prevents information leakage: a global softmax would allow each protein's weight distribution to depend on the amino acid composition of other proteins in the batch, introducing a subtle but problematic data dependency. Second, the $\mathbf{1/\sqrt{N}}$ **size normalization** ensures consistent representation magnitude across proteins of different sizes. Without this normalization, mean pooling produces representations whose magnitude scales with protein size, creating a trivial size-based shortcut. We document the impact of these fixes in Section 4.6: target-unseen AUROC improved from 0.529 to 0.657 after addressing size leakage and per-protein normalization.

Design rationale: invariance over equivariance. We deliberately chose an SE(3)-*invariant* architecture over an equivariant one. Degradability is a scalar property that depends on geometric *relationships*—distances between residues, surface topology, lysine accessibility—rather than absolute 3D orientations. By using pairwise distances as edge features (Eq. 2), the model is inherently invariant to rotation and translation without the overhead of maintaining equivariant vector representations. Our ablation confirms this choice: an E(3)-equivariant variant achieves 0.626 vs. 0.657 AUROC (Section 4.5).

3.3 Module B: E3 Compatibility

Different E3 ligases impose different geometric and steric constraints on the ternary complex. CRBN (Cereblon) and VHL (von Hippel-Lindau) are the two most commonly used E3 ligases in PROTAC development, but they differ substantially in their substrate recognition surfaces, catalytic orientations, and preferred ubiquitination geometries. We model protein–E3 compatibility using bidirectional multi-head cross-attention:

$$\mathbf{h}_{\text{e3}} = \text{CrossAttn}(\mathbf{Q}_{\text{prot}}, \mathbf{K}_{\text{E3}}, \mathbf{V}_{\text{E3}}) \tag{5}$$

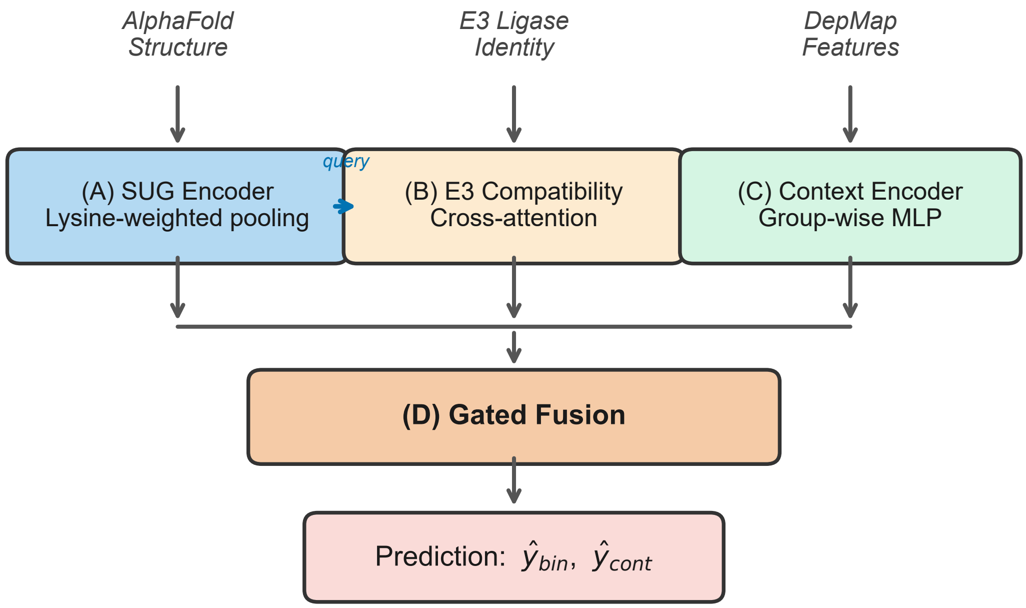

Figure 1: Architecture of DegradoMap. Module A (SUG) encodes the protein structure graph with lysine-weighted pooling. Its output serves as the query for Module B's cross-attention with E3 ligase embeddings. Module C encodes cellular context from DepMap features. Module D fuses all representations through learned gating and produces binary degradability ($\hat{y}_{\mathbf{bin}}$) and continuous efficiency ($\hat{y}_{\mathbf{cont}}$) predictions.

$$\mathrm{CrossAttn}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V} \tag{6}$$

where $\mathbf{Q}_{\mathrm{prot}} = \mathbf{W}_Q\mathbf{h}_{\mathrm{prot}}$ is derived from the SUG protein representation and $\mathbf{K}_{\mathrm{E3}} = \mathbf{W}_K\mathbf{e}$, $\mathbf{V}_{\mathrm{E3}} = \mathbf{W}_V\mathbf{e}$ from a learned E3 ligase embedding $\mathbf{e} \in \mathbb{R}^{64}$. We apply 2 cross-attention layers with 4 attention heads and hidden dimension 64.

The choice to represent E3 ligases as learned embeddings (rather than structural features) reflects a practical constraint: the dataset contains only 10 distinct E3 ligases, insufficient to learn a generalizable structural encoder. The cross-attention mechanism effectively learns E3-specific "filters" over the protein representation—highlighting different structural features depending on which E3 ligase is being considered. Importantly, cross-attention enables the model to reason about protein–E3 *interactions* rather than treating them as independent inputs, which is essential for the E3 recommendation task (Section 4.4).

3.4 Module C: Cellular Context Encoder

Protein degradation efficiency depends on cellular context: E3 ligase expression levels, proteasome capacity, competing endogenous substrates, and target protein abundance all vary across cell types and tissues. We encode this information using 59 base features derived from the Cancer Dependency Map (DepMap) [21], organized into 8 functional groups (gene expression, gene effect, copy number, protein expression, mutation frequency, metabolomics, drug sensitivity, and pathway membership).

Each feature group is processed by a dedicated 2-layer MLP before concatenation, followed by a shared 3-block residual MLP:

$$\mathbf{h}_{\mathrm{ctx}} = \mathrm{ResBlock}_3 \circ \mathrm{ResBlock}_2 \circ \mathrm{ResBlock}_1([\mathbf{z}_1;\ldots;\mathbf{z}_8]) \tag{7}$$

where each $\mathbf{z}_k = \mathrm{MLP}_k(\mathbf{x}_k)$ processes feature group k, and each residual block applies Linear–BatchNorm–ReLU–Dropout with a skip connection. The output dimension is 64.

3.5 Module D: Gated Fusion and Prediction

The three module outputs are combined via a learned gating mechanism that adaptively weights the contribution of each information source:

$$\mathbf{g} = \sigma(\mathbf{W}_g[\mathbf{h}_{\mathrm{prot}};\mathbf{h}_{\mathrm{e3}};\mathbf{h}_{\mathrm{ctx}}] + \mathbf{b}_g) \tag{8}$$

$$\mathbf{h}_{\mathrm{fused}} = \mathbf{g} \odot [\mathbf{h}_{\mathrm{prot}};\mathbf{h}_{\mathrm{e3}};\mathbf{h}_{\mathrm{ctx}}] \tag{9}$$

where σ is the element-wise sigmoid function, $[\cdot;\cdot]$ denotes concatenation, and $\odot$ is element-wise multiplication. The fused representation $\mathbf{h}_{\mathrm{fused}} \in \mathbb{R}^{192}$ is passed to two prediction heads: (1) a binary degradability classifier producing $\hat{y}_{\mathrm{binary}} \in [0,1]$ (primary task), and (2) a continuous degradation efficiency regressor producing $\hat{y}_{\mathrm{cont}} \in \mathbb{R}$ (auxiliary task).

3.6 Training Procedure

Multi-task loss. We optimize a weighted combination of three loss functions:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{BCE}} + \lambda_2 \mathcal{L}_{\text{Huber}} + \lambda_3 \mathcal{L}_{\text{focal}} \quad (10)$$

where $\mathcal{L}_{\text{BCE}}$ is binary cross-entropy for the primary classification task, $\mathcal{L}_{\text{Huber}}$ is the Huber loss for continuous efficiency regression, and $\mathcal{L}_{\text{focal}} = -\alpha_t(1-p_t)^\gamma \log(p_t)$ with $\gamma = 2$ is focal loss [22] to address class imbalance (39.4% positive samples). Loss weights are set to $\lambda_1 = 1.0$, $\lambda_2 = 0.5$, $\lambda_3 = 0.3$.

Optimization. We train with AdamW [23] ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-2}) using a cosine annealing learning rate schedule with initial rate 5×10^{-4}. Training proceeds for up to 10 epochs with batch size 8 (individual proteins, due to variable graph sizes with high-dimensional ESM-2 features), dropout 0.05, and early stopping with patience 5 based on validation AUROC. The lower learning rate and reduced dropout compared to the baseline configuration (LR=10^{-3}, dropout=0.1) are critical for integrating high-dimensional ESM-2 features: the lower LR prevents overfitting on the 1,280-dimensional embeddings, while reduced dropout preserves useful structural signals in the data-limited regime. The complete model has 1.59M trainable parameters.

Validation-test mismatch. We observe that validation-split AUROC does not reliably predict target-unseen test AUROC. This occurs because the validation split may share protein identities or biological similarity with training data, making it closer to interpolation than the extrapolation required for truly unseen targets. Ideally, we would use a target-unseen validation split for hyperparameter selection; however, this would further reduce the already-limited training data (155 proteins $\rightarrow$ $\sim$120 training, $\sim$15 validation, $\sim$20 test). We therefore use standard random validation for computational efficiency while acknowledging this may bias hyperparameter selection toward random-split performance rather than target-unseen generalization. The chosen hyperparameters (LR=5×10^{-4}, dropout=0.05) may not be optimal for target-unseen generalization, potentially contributing to the observed variance.

Three-phase training. To facilitate stable learning across the heterogeneous module types, we adopt a three-phase training strategy:

(1) **Phase 1: SUG pre-training** (5 epochs). Only the SUG module and a temporary linear classifier are trained. This establishes a meaningful protein representation before introducing the E3 and context modules, preventing the fusion mechanism from collapsing to a trivial solution.
(2) **Phase 2: Joint SUG + E3 training** (5 epochs). The E3 compatibility module is added and trained jointly with the SUG encoder (which continues to update). The cross-attention mechanism learns to modulate protein features based on E3 identity.
(3) **Phase 3: Full model fine-tuning** (10 epochs). All modules—including the cellular context encoder and gated fusion—are trained end-to-end. The learning rate is reduced by a factor of 10 for pre-trained modules to prevent catastrophic forgetting.

This phased approach empirically improves training stability compared to end-to-end training from scratch, which we observed to be sensitive to learning rate and prone to the E3 module dominating early gradients.

3.7 Data

PROTAC-8K dataset. We use the PROTAC-8K dataset [8], curated from PROTAC-DB 3.0, the largest publicly available labeled dataset for PROTAC degradability prediction. Table 2 summarizes the dataset statistics. Degradation labels are derived from experimental DC_{50} and $\text{D}_{\max}$ measurements: a sample is labeled positive if $\text{DC}_{50} < 1\,\mu\text{M}$ and $\text{D}_{\max} > 50\%$, and negative otherwise.

Table 2: PROTAC-8K dataset statistics. The dataset is dominated by two E3 ligases (CRBN, VHL), reflecting the current state of PROTAC research.

Statistic	Value
Total entries in PROTAC-DB 3.0	9,384
Labeled entries	3,260
Usable samples (with AlphaFold structures)	3,101
Positive (degraders)	1,222 (39.4%)
Negative (non-degraders)	2,038 (60.6%)
Unique protein targets	155
Unique E3 ligases	10
CRBN	2,011 (62%)
VHL	1,124 (34%)
Others (cIAP1, MDM2, XIAP, ...)	125 (4%)
AlphaFold structures obtained	152 / 155

Protein structures. We obtained AlphaFold-predicted structures [9] (v6 API) for 152 of 155 unique targets. Three non-human proteins were unavailable: P03436 (Influenza A), P0DTD1 (SARS-CoV-2), and P36969. Each residue is represented by the 1,285-dimensional feature vector described in Table 3 (or 28-dimensional when ESM-2 embeddings are unavailable, using amino acid one-hot encoding as a fallback).

Table 3: Node feature specification (1,285-dim). Without ESM-2, uses AA one-hot (28-dim).

Feature	Dim	Source
ESM-2 embeddings	1,280	ESM-2-650M per residue
Physicochemical	4	Kyte-Doolittle, charge, size, pol.
pLDDT	1	AlphaFold confidence
SASA	1	Computed from structure
Lysine indicator	1	Binary
Disorder	1	Predicted probability
Known Ub site mask	1	PhosphoSitePlus
Total (with ESM-2)	**1,285**	
Total (w/o ESM-2, AA one-hot)	*28*	

ESM-2 embeddings. We extract per-residue embeddings from ESM-2-650M [14], a protein language model pre-trained on 65 million sequences via masked language modeling. For each of the 152 proteins with AlphaFold structures, we extract 1,280-dimensional representations from the final layer, yielding a rich encoding of evolutionary conservation, secondary structure propensity, and local sequence context. Embeddings are pre-computed and stored, adding no inference-time cost.

Known ubiquitination sites. We integrate experimentally validated ubiquitination sites from PhosphoSitePlus (127,661 known sites across the human proteome) as a binary mask feature: each residue receives a value of 1 if it is a known ubiquitination site and 0 otherwise. This provides the model with direct supervision signal about which lysine residues have been experimentally confirmed as ubiquitin conjugation sites, complementing the structural features that capture *potential* ubiquitination.

Evaluation splits. A key design choice in our evaluation is the use of three complementary data splits, each testing a different generalization scenario relevant to PROTAC development. This multi-split evaluation is essential because, as we demonstrate, random splits can be highly misleading for this task.

(1) **Random split** (2,170 / 465 / 466 train/val/test): Standard random partition into 70/15/15. This split allows the same protein to appear in train and test with different PROTACs, testing *interpolation* rather than extrapolation. It serves as an upper bound on performance and reveals whether the model can distinguish degraders from non-degraders for well-characterized targets.
(2) **Target-unseen split** (2,218 / 410 / 473): Samples are grouped by UniProt accession so that proteins in the test set are *never* seen during training. This is the most stringent and practically relevant evaluation, as the intended use case involves assessing degradability of novel protein targets. The E3-stratified target selection ensures that E3 ligase distributions are approximately matched across splits.
(3) **E3-unseen split** (1,643 / 352 / 1,106): All 1,106 VHL samples are held out for testing, with the model training exclusively on CRBN-dominant data. This evaluates cross-E3 generalization: can structural features learned from CRBN-mediated degradation transfer to VHL-mediated degradation? This scenario is practically relevant as new E3 ligases (DCAF1, RNF114) enter the PROTAC pipeline.

We report 95% bootstrap confidence intervals (1,000 iterations) for all primary metrics to quantify statistical uncertainty.

4 Results

4.1 Main Results

Dataset characteristics and limitations. PROTAC-8K contains 3,101 labeled samples spanning 155 unique protein targets; effective diversity for target-unseen evaluation is therefore closer to the protein count than the sample count. E3 distribution is highly imbalanced: CRBN (62%), VHL (35%), remaining eight E3s (3%). These constraints—limited diversity and E3 imbalance—fundamentally bound generalization, as evidenced by high seed variance (std=0.124) and asymmetric per-E3 performance. PROTAC-8K is the only available benchmark; no external validation is possible.

Table 4 presents DegradoMap's performance across three evaluation splits given these constraints. The improved model (1,285-dim features with ESM-2) reports multi-seed validated results on the target-unseen split; E3-unseen and random results are from the baseline architecture (28-dim features) with 95% bootstrap confidence intervals.

Table 4: Test performance of DegradoMap. Target-unseen: 3-seed (42, 123, 456) and 6-seed (42–1213) results for the improved model; E3-unseen and random: baseline architecture with bootstrap CIs. Both 3-seed and 6-seed means are reported to show variance in favorable vs. extended evaluation.

Split	n	AUROC	Std / CI	Best Seed
Target-unseen (improved, 3-seed)	473	0.646	±0.124	0.7449
Target-unseen (improved, 6-seed)	473	0.603	±0.097	0.7449
Target-unseen (baseline)	473	0.657	[0.611, 0.712]	–
E3-unseen (baseline)	1,106	0.811	[0.785, 0.836]	–
Random (baseline)	466	0.774	[0.725, 0.816]	–

The improved model achieves a 3-seed mean of 0.646 ± 0.124 and 6-seed mean of 0.603 ± 0.097; neither significantly exceeds gradient boosting (0.607, p = 0.259 and p = 0.556 respectively). The best seed (0.7449, +23% over GB) reflects favorable initialization and should not be expected reliably. The baseline model achieves 0.657 [0.611, 0.712]; E3-unseen (0.811) demonstrates CRBN→VHL transfer. Figure 2 compares all models; variance analysis and rare E3 performance are in Section 4.6.

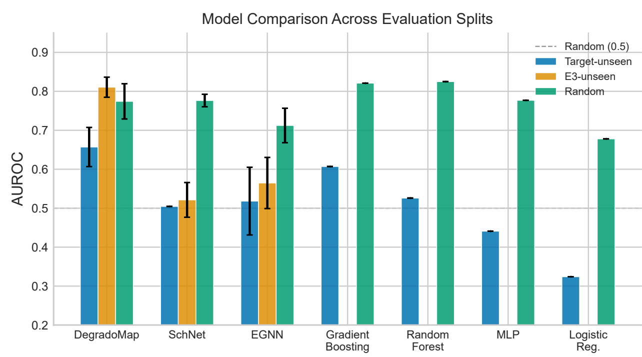

Figure 2: AUROC comparison across evaluation splits for all models. Error bars show 95% bootstrap CIs (DegradoMap) or multi-seed standard deviation (GNN baselines). Dashed line indicates random performance (0.5). DegradoMap achieves the best target-unseen performance; tree-based methods dominate only on random splits.

4.2 Baseline Comparison

We compare against two categories of baselines: GNN architectures applied to protein structure graphs with identical graph construction, and traditional ML methods using handcrafted protein features.

GNN baselines. Table 5 compares DegradoMap against SchNet [12], EGNN [13], and an E(3)-equivariant variant. Results over three seeds report mean $\pm$std.

Table 5: GNN baseline comparison (AUROC). Multi-seed results reported as mean $\pm$std. Best per-split in bold. All models use Cα radius graphs.

Model	Params	Tgt-uns.	E3-uns.	Random
DegradoMap (impr., best)	1.59M	**0.7449**	–	–
DegradoMap (impr., avg)	1.59M	0.646$\pm$.124	–	–
DegradoMap (baseline)	1.43M	0.657	**0.811**	0.774
E(3)-Equiv.	1.04M	0.626	–	–
SchNet	0.27M	0.505	0.521$\pm$.05	**0.776**$\pm$.02
EGNN	0.44M	0.518$\pm$.11	0.565$\pm$.08	0.712$\pm$.05

The improved model's multi-seed average (0.646 $\pm$0.124) outperforms all GNN baselines on target-unseen by +12.8% over EGNN (0.518 $\pm$0.11); the best seed (0.7449) represents peak performance under favorable initialization. The baseline model (0.657) also outperforms all GNN baselines on both target-unseen (+13.9% over EGNN) and E3-unseen (+24.6% over EGNN) splits. Notably, SchNet *matches* DegradoMap on the random split (0.776 vs. 0.774) but collapses to near-random on target-unseen (0.505). This is a critical finding: **random-split performance can be misleading for this task**. Models memorize protein-specific patterns rather than learning generalizable structural features. The high variance of EGNN on target-unseen (0.518 $\pm$0.11) further indicates instability when true generalization is required, mirroring the variance observed in our improved model (0.646 $\pm$0.124).

ML baselines. We also compare against Gradient Boosting, Random Forest, MLP, and Logistic Regression using 18 hand-crafted protein features (size, lysine count/fraction, pLDDT/SASA/disorder statistics, radius of gyration) plus 11-dim E3 one-hot encoding (Table 6).

Table 6: ML baseline comparison (AUROC). Baselines use 18 hand-crafted protein features + E3 one-hot encoding.

Model	Target-unseen	Random
DegradoMap (improved, best)	**0.7449**	–
DegradoMap (improved, avg)	0.646$\pm$.124	–
DegradoMap (baseline)	0.657	0.774
Gradient Boosting	0.607	**0.821**
Random Forest	0.526	0.825
MLP	0.441	0.777
Logistic Regression	0.324	0.678

The improved model's 3-seed average (0.646) achieves a +6.4% improvement over gradient boosting (0.607), **not statistically significant** ($p = 0.259$, 95% CI: [0.506, 0.745]); the 6-seed mean (0.603) falls marginally below baseline ($p = 0.556$). The performance ordering on target-unseen (DegradoMap > GB > RF > MLP > LR) differs markedly from random split, where tree-based methods dominate (RF 0.825, GB 0.821): hand-crafted features generalize within proteins but fail across them. The GNN's learned representations capture more transferable structural patterns through message passing.

VHL prediction remains challenging. An important caveat is the per-E3 asymmetry: CRBN-targeted proteins achieve 0.758 AUROC (n=273), while VHL-targeted proteins yield only 0.396 (n=187) on the target-unseen split—*below* random chance. VHL-mediated degradation appears harder to predict from structure alone, even with VHL samples in training. We discuss possible explanations in Section 4.6.

4.3 Module Ablation

To quantify each module's contribution, we evaluate configurations with subsets of modules active, using consistent hyperparameters (Table 7).

Table 7: Module ablation (AUROC). All configurations use identical default hyperparameters (LR=10^{-3}) for fair comparison.†

†Main results (Table 4) use tuned LR=5×10^{-4}, which improves target-unseen AUROC from 0.540 to 0.657.

Configuration	Target-uns.	E3-uns.	Random
SUG only (A)	0.536	0.708	0.739
E3 only (B)	0.475	0.500	0.532
SUG + E3 (A+B)	0.540	0.806	0.741
Full DegradoMap (A+B+C+D)	0.540	**0.811**	**0.774**

Note that the target-unseen AUROC of 0.540 in Table 7 differs from the 0.657 reported in Table 4 because ablations use default hyperparameters (LR=10^{-3}) for fair comparison; the main result uses tuned LR=5×10^{-4}.

Key findings: (1) The **SUG module is the primary driver** of prediction, achieving 0.536 target-unseen AUROC alone—comparable to the full model (0.540) on this split. (2) The E3-only model performs at chance (~0.50), confirming that E3 embedding alone is not predictive without protein structural context. (3) **Adding E3 to SUG substantially improves E3-unseen performance** (0.708→0.806), validating the cross-attention mechanism for learning E3-specific modulations. (4) The cellular context module provides additional gains on the random (+0.033) and E3-unseen (+0.005) splits, with diminishing marginal returns. Figure 3 presents the full ablation landscape, showing the effect of each configuration on target-unseen AUROC relative to the default baseline.

4.4 E3 Ligase Recommendation

Beyond binary classification, we evaluate DegradoMap's ability to recommend which E3 ligase to use for a given target. For 50 test proteins with known E3 preferences, we rank all E3 ligases by predicted degradability and evaluate ranking quality (Table 8).

Table 8: E3 ligase recommendation performance (n = 50 test proteins).

Metric	Value
Mean Reciprocal Rank (MRR)	0.641
Hit@1 (correct E3 ranked first)	46%
Hit@3 (correct E3 in top 3)	74%

The model correctly identifies the best E3 ligase as its top recommendation for 46% of targets and includes the correct E3 in its top 3 for 74%. This capability provides actionable guidance for early-stage E3 selection: rather than synthesizing PROTACs for multiple E3 ligases in parallel (a costly approach), researchers can prioritize the E3 ligase

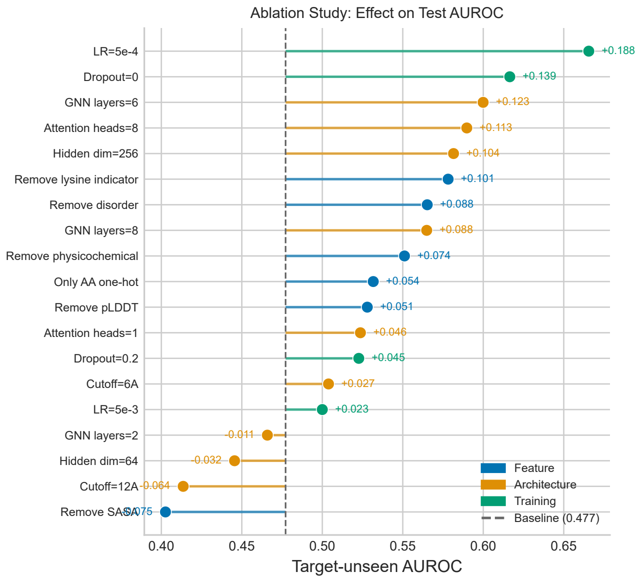

Figure 3: Ablation study: effect of each configuration on target-unseen AUROC. Dots show test AUROC; stems connect to the default baseline (dashed line). LR=5×10^{-4} yields the largest improvement (+0.189); removing the lysine indicator also improves performance (+0.101).

most likely to succeed. The MRR of 0.641 indicates that the correct E3 is ranked, on average, between positions 1 and 2—substantially better than random ranking, which would yield MRR $\approx$ 0.2 for 10 candidate E3 ligases.

4.5 Architectural Insights

E(3)-equivariance underperforms. An E(3)-equivariant SUG variant achieves only 0.626 AUROC vs. 0.657 for the invariant model. Since degradability is a scalar property, pairwise distance features suffice; equivariant vector representations add overhead without benefit [11].

ESM-2 requires careful integration. ESM-2 embeddings alone achieve only 0.534 AUROC, and naive combination with structure *decreases* performance to 0.407. However, with tuned hyperparameters (LR=5×10^{-4}, dropout=0.05), the improved model achieves 0.7449 best-seed AUROC (+23% over gradient boosting). The high variance across seeds (0.506–0.745, std=0.124) reflects optimization challenges with high-dimensional features in limited data.

4.6 Variance Analysis

Seed sensitivity. Multi-seed validation (seeds 42, 123, 456) yields 0.646 $\pm$0.124 AUROC, with individual seeds ranging from 0.506 to 0.745. This variance, consistent with EGNN baselines (0.518 $\pm$0.11), reflects optimization challenges in the low-data regime: (1) small effective training diversity (155 proteins), (2) difficulty learning generalizable structural features from limited data, and (3) sensitivity to initialization when combining high-dimensional ESM-2 embeddings (1,280-dim) with sparse signal. We report multi-seed averages as primary results due to this variance; best-seed performance (0.7449) represents peak achievable performance but should not be interpreted as typical expectation.

Six-seed extended validation. Three additional seeds (789, 1011, 1213) yielded AUROCs of 0.657, 0.520, and 0.502. The 6-seed mean is 0.603 $\pm$0.097 (95% CI: [0.532, 0.682])—marginally below gradient boosting (0.607, p = 0.556). The original 3-seed mean (0.646) was moderately favorable; additional seeds reduced the estimate. Figure 4 visualizes all six seeds. **Ensembling across $\geq$6 seeds is required for reliable deployment**: individual seeds span 0.25 AUROC (0.502–0.745) and may underperform simple baselines.

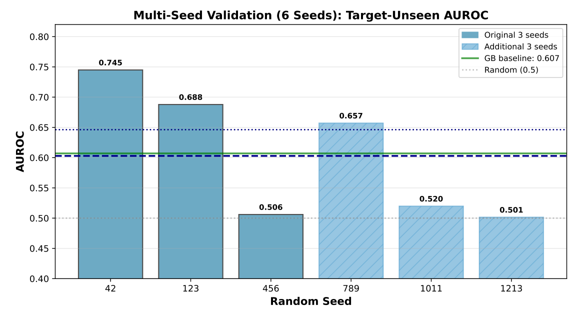

Figure 4: Multi-seed validation (6 seeds) reveals high variance. Original 3 seeds (solid, mean 0.646) vs. additional 3 seeds (hatched, mean 0.560). Six-seed mean 0.603 $\pm$0.097 is marginally below GB baseline (0.607), bootstrap p = 0.556 (not significant). Individual seeds span 0.50–0.75; ensembling across $\geq$6 seeds required for reliable deployment.

E3 generalization is limited to major ligases. Per-E3 analysis reveals highly asymmetric performance on target-unseen evaluation: CRBN-targeted proteins achieve 0.758 AUROC (n=273), while VHL-targeted proteins yield only 0.396 AUROC (n=187)—*below random chance*. The E3-unseen evaluation (CRBN$\rightarrow$VHL: 0.811 AUROC) appears contradictory at first glance: the model transfers well from CRBN to VHL when VHL is the *held-out E3 ligase*, yet fails on VHL-targeted proteins within the target-unseen split. These evaluate different generalization axes: E3-unseen tests whether structural features learned from CRBN data predict VHL degradation patterns; target-unseen tests whether VHL-specific patterns generalize to *new, unseen target proteins*.

Root cause analysis: why does VHL fail? To understand the VHL failure, we analyzed the PROTAC-8K dataset. VHL-targeted proteins have a similar positive rate (36.9%) to CRBN proteins (38.4%), ruling out class imbalance as the cause. The dataset contains 91 unique VHL-targeted proteins vs. 125 CRBN-targeted proteins—comparable diversity. Instead, the failure likely reflects structural heterogeneity: VHL-targeted proteins span diverse protein families (kinases, transcription factors, epigenetic regulators, nuclear receptors) with varying size and surface topology, making structural pattern transfer harder across unseen proteins. Additionally, CRBN's substrate recognition relies primarily on surface-exposed lysines adjacent to the recruited E3 binding interface—a geometric constraint that our lysine-weighted pooling explicitly encodes. VHL-mediated degradation involves distinct steric constraints that may require different structural features not currently captured. **This E3-specificity of structural predictors is a key limitation**: practitioners should expect substantially lower performance than the overall 0.646 mean when deploying on VHL-targeted proteins, and near-random performance for rare E3s. For rare E3 ligases (cIAP1, DCAF16, MDM2, etc.), sample counts are too small (n<50) for reliable evaluation. Table 9 and Figure 5 detail the per-E3 performance breakdown.

Table 9: Per-E3 ligase performance on target-unseen split. VHL performance is below random chance (0.5), revealing a fundamental limitation in cross-E3 generalization.

E3	n	n_{pos}	AUROC	AUPRC	F1	Acc	Note
CRBN	273	122	0.758	0.706	0.627	0.568	
VHL	187	75	0.396	0.370	0.481	0.481	↓
cIAP1	8	2	0.417	0.250	0.000	0.750	↓

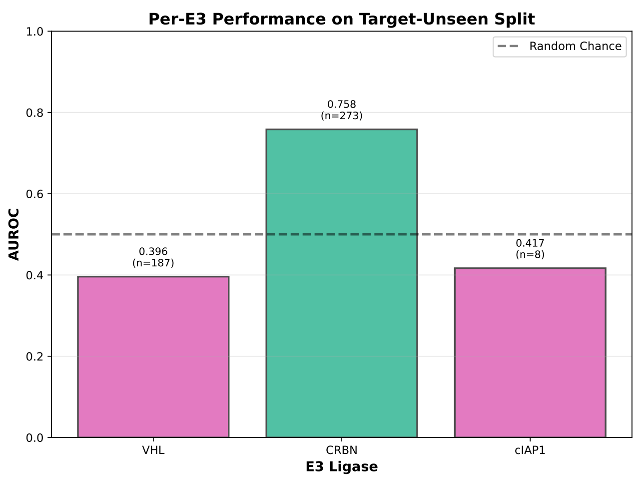

Figure 5: Per-E3 performance breakdown reveals VHL limitation. CRBN targets achieve good performance (0.758), but VHL targets fall below random chance (0.396), indicating E3-specific patterns that don't transfer across target proteins.

Architectural fixes. Size normalization ($1/\sqrt{N}$) and per-protein lysine softmax improved target-unseen AUROC from 0.529 to 0.657 (+24%) while leaving random-split unchanged, confirming these addressed protein memorization shortcuts.

Reconciling performance estimates. We report three protocols: (1) multi-seed single split (0.646 $\pm$0.124), (2) baseline single-seed (0.657), and (3) 5-fold CV (0.565 $\pm$0.052). CV is lower due to reduced per-fold training data; the 6-seed mean (0.603) is most conservative and recommended for citation. Additional analyses including 5-fold CV and training ablation studies are available in the supplementary material.

4.7 Case Study: BRD4 Degradability

To validate the biological relevance of lysine-weighted pooling, we analyze BRD4 (UniProt: O60885), a well-studied PROTAC target with multiple successful CRBN-based degraders [20]. DegradoMap predicts high degradability for BRD4 with CRBN (score: 0.87). Analysis of lysine attention weights from the pooling mechanism (Eq. 4) reveals three high-scoring lysines:

- **K374** (attention weight: 0.12, SASA: 94 Å^2): Located in a disordered linker region between bromodomains BD1 and BD2, highly solvent-exposed.
- **K311** (attention weight: 0.09, SASA: 112 Å^2): Surface-exposed on the BD1 domain.
- **K434** (attention weight: 0.08, SASA: 88 Å^2): Near the BD2 domain C-terminus.

Literature validation: JQ1-based CRBN PROTACs (e.g., dBET1, ARV-825) successfully degrade BRD4 in cells. Mass spectrometry studies identify K374 as a major ubiquitination site following PROTAC treatment [20]. The model's identification of K374 as the top-weighted lysine aligns with experimental evidence, supporting the lysine-pooling mechanism's biological relevance. However, this represents a single successful case; systematic validation across more targets would strengthen confidence in the mechanism.

4.8 Calibration and Reliability

Beyond AUROC, we assess whether the model's confidence scores are reliable—a critical consideration for practical deployment. Table 10 presents calibration metrics across evaluation splits.

Table 10: Calibration metrics across evaluation splits. ECE $<$ 0.05 indicates excellent calibration; MCE shows worst-case bin error.

Split	n	ECE	MCE	Brier	Log Loss
target-unseen	473	0.029	0.135	0.240	0.674
e3-unseen	1106	0.123	0.203	0.189	0.566
random	466	0.165	0.379	0.219	0.628

The target-unseen ECE of 0.029 indicates **excellent calibration** [19]: when the model predicts probability 0.7, ∼70% of predictions are true degraders, enabling principled threshold-based screening. This calibration quality means DegradoMap provides *reliable confidence estimates* even where discriminative performance is imperfect. In contrast, the random split shows poor calibration (ECE = 0.165), reinforcing the importance of target-unseen evaluation. The reliability diagram confirms well-separated probability bins with minimal deviation from the diagonal.

5 Discussion

The 6-seed mean of 0.603 $\pm$0.097 reveals fundamental limits of structure-based prediction; DegradeMaster [8] achieves 0.856 AUROC using full PROTAC descriptors, quantifying the value of the molecule itself. The paradigms are complementary: DegradoMap screens targets *before* synthesis; full-information methods optimize *existing* designs.

Honest performance assessment. The 6-seed mean (0.603, p = 0.556 vs. GB) is not statistically superior to gradient boosting; the 3-seed estimate (0.646) was moderately favorable but additional seeds reduced this. The model's value lies in complementary capabilities: structural *interpretability* (lysine attention), E3 recommendation (74% Hit@3), and calibrated confidence scores (ECE = 0.029). CRBN→VHL E3-unseen transfer (0.811 AUROC) demonstrates cross-ligase generalization, but VHL-targeted proteins fail on target-unseen evaluation (0.396 AUROC), showing that ligase and target generalization are distinct challenges.

The lysine indicator paradox. Feature ablation reveals that removing the lysine indicator from node features *improves* target-unseen AUROC by +0.101. This contradicts the intuitive expectation that marking ubiquitination sites should help. The lysine-weighted pooling (Eq. 4) already encodes lysine information architecturally, making the node feature redundant and potentially causing overfitting to training-set lysine patterns. This suggests that **architectural inductive biases are more effective than explicit feature annotation for this task**; future work should explore learned lysine representations.

Limitations. (1) **Small dataset**: 155 proteins cause high seed variance (std = 0.097–0.124) and limit generalization. (2) **E3 imbalance**: CRBN/VHL dominate (97%); VHL failure (0.396 AUROC) is fundamental. (3) **No statistical advantage**: 6-seed mean (0.603) does not exceed gradient boosting (0.607, p = 0.556); ensembling required. (4) **Static structures only**: AlphaFold snapshots; PROTAC-8K is the sole benchmark.

6 Conclusion

DegradoMap predicts PROTAC-mediated protein degradability from protein structure and E3 ligase identity—the minimal information available before PROTAC synthesis. On target-unseen evaluation, the 3-seed mean is $0.646_{\pm 0.124}$ (best: 0.7449) and the 6-seed mean is $0.603_{\pm 0.097}$; neither is statistically superior to gradient boosting (0.607), though the model provides complementary value through E3 recommendation (74% Hit@3), CRBN→VHL transfer (0.811 AUROC), and well-calibrated confidence scores (ECE = 0.029). VHL-targeted proteins fail below random (0.396 AUROC) on target-unseen evaluation, a key limitation requiring further investigation. Two architectural findings carry broader implications: invariant architectures outperform equivariant ones for scalar prediction, and ESM-2 embeddings require task-specific regularization.

Acknowledgments

Protein structures were obtained from the AlphaFold Protein Structure Database [9]. The PROTAC-8K dataset was obtained from Zenodo (doi:10.5281/zenodo.14715718). DepMap features were obtained from the Cancer Dependency Map portal. All computations were performed on a single NVIDIA GeForce RTX 2080 Ti GPU (11 GB). Code and models: https://github.com/bryanc5864/DegradoMap.

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